Under the RAM model, the performance of an algorithm is measured as the number of steps it takes on a given input.

These simplifications allow us to abstract out the details of the platform. For some problems, or some variants, we might be interested in bringing some details back. For instance, we can distinguish cache accesses from disk accesses. But in general, this generalization serves us well.

Now that we've established the model we'll use, let's discuss what we'll consider a relevant improvement of an algorithm.

B.3 Order of Magnitude

The other aspect we still need to simplify is the way we count instructions.

We might decide to count only some kind of operations like memory accesses. Or, when analyzing sorting algorithms, we can only take into account the number of elements swaps.